

End User Slides

TGIR QuantLib Tools

TG Investments and Research

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What Users Need To Know

- Login at `http://127.0.0.1:5050`.
- Main workflow lives on `/dashboard`.
- Five trades are always visible on the blotter.
- The top bar keeps nav, research, GitHub, export, realtime, and reset controls together.
- Detailed trade changes happen in the dedicated editor pages.

Dashboard Components

- Valuation date field shared by all rates analytics.
- SOFR curve inputs plus a compact market-rate / zero-rate table.
- Market-versus-zero-rate chart plus mean reversion shown as a percentage.
- Workbook ATM swaption normal-vol matrix.
- Bermudan pricing grid.
- On-demand forward-strip and OIS-repricing diagnostics.
- SPX cliquet market assumptions.

Trade Editors

- Swap: direction, notional, fixed coupon, tenor.
- European swaption: strike, expiry, underlying tenor, then Hull-White 1F + Jamshidian pricing.
- The four rates trades start at 100,000,000.00 notionals by default.
- Bermudan swaption editors: first exercise, fixed final maturity, pay/reset frequency, model choice.
- Bermudan detail view shows the exercise-to-swap mapping, helper source points, and Hull-White versus G2++ comparison.
- Equity cliquet: option type, quantity, moneyness, maturity, reset frequency.

Cliquet Page Outputs

- Analytic Greeks from QuantLib.
- Reset-by-reset forward-start decomposition.
- Spot-volatility scenario matrix.
- Monte Carlo payoff distribution summary.

Exports And Reference Pages

- Download `/curve-debug.csv` for audit or spreadsheet work.
- Use `/quantlib-data-model` for the full model map.
- The model page also lists the swaption and cliquet research papers.

Troubleshooting

- Login issues usually mean the local `.env` credentials are wrong.
- Bermudan risk is slow because the model recalibrates on each relevant bump.
- Cliquet rendering is slower than the other trades because it includes scenarios and Monte Carlo analytics.
- Use the reset button to restore the demo state.